# Project name: Enhancing Operational Efficiency in a Multi-Specialty Hospital

| **Project Overview** |
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| This project aims to improve hospital operational efficiency by enhancing appointment scheduling, resource allocation, and inter-departmental communication through an integrated Hospital Information System (HIS). |

| **Background and Problem Statement** |
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| The hospital currently faces challenges in appointment scheduling, resource management, and communication across departments. Patients experience long wait times and unclear communication, while staff struggle with overbooking, outdated systems, and limited system integration. |

| **Project scope** | |
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| **In-scope** | **Out-of-scope** |
| 1. Integration of scheduling and record management systems<br>2. Real-time appointment notifications via email/SMS<br>3. Improved visibility of resource availability<br>**4.** Data analytics for patient and resource trends | 1. Replacement of all clinical equipment<br>2. Redesign of hospital physical infrastructure |

| **Stakeholders** |
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| Primary stakeholders include Patients, Doctors, Nurses, Administrative Staff, and IT Teams. Each group experiences specific challenges related to scheduling, resource availability, and system usability. |

| Business objectives |
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| 1. Reduce average patient waiting time by at least 30% within 3 months of system implementation.<br>2. Achieve ≥95% successful online bookings with zero double bookings within the first release.<br>3. Increase doctor resource utilization to 70–85% within 3 months through optimized scheduling.<br>4. Enable real-time data integration across all departments (admin, clinical, pharmacy, billing) at go-live.<br>5. Improve patient satisfaction score by at least 20% within 6 months after implementation.<br>6. Maintain system uptime of ≥99.5% while ensuring full regulatory and data security compliance. |

| Requirements | |
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| **Functional requirements** | **Non-functional requirements** |
| 1. Online appointment booking with real-time availability<br>2. Automated appointment reminders and notifications<br>3. Integrated patient records accessible across departments<br>4. Resource dashboard for administrators to monitor operations including rooms, equipment, and staff | 1. System uptime of at least 99.5%<br>2. Secure data storage compliant with healthcare regulations<br>3. User-friendly interfaces for patients and staff<br>4. Scalability to support future growth |

| Assumptions |
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| 1. Hospital management actively supports system integration and change initiatives<br>2. Clinical and administrative staff will receive adequate training and adopt the new system<br>3. Existing patient and operational data are sufficiently accurate for system integration<br>4. Patients have access to basic digital tools and are willing to use online services<br>5. Interdepartmental processes can be standardized across units |

| Constraints |
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| 1. Limited IT budget requiring phased implementation<br>2. Dependency on existing legacy systems and infrastructure<br>3. Regulatory, data privacy, and healthcare compliance requirements<br>4. Limited availability of staff time for training and change activities<br>5. Fixed project timeline aligned with hospital operational priorities |

| Supporting Data |
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| Appointment Records<br>Patient Feedback Entries<br>Resource Utilization Records |

| Conclusion |
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| Implementing an integrated Hospital Information System is expected to significantly improve scheduling accuracy, resource utilization, staff efficiency, and overall patient satisfaction. |